

COURSE DD2380 ARTIFICIAL INTELLIGENCE

ETHICAL AND SOCIETAL ASPECTS OF AI



Group members: Nor Hanchi, Judy Florence, Mo Kong, Bahar Kimanos

Group number: 9

Nor Hanchi, nhanchi@kth.se,

Text: 15, 16

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In the article: "Ethical Issues in Advanced Artificial Intelligence" Bostrom (2026) makes a simple case for an increased preparedness in the coming of so-called super intelligences. Here it is differentiated from general AI from its supposed ability to exceed human intelligence in various domains of existence, giving rise to advanced technologies that can further reach into non-matured areas such as space travel and von Neumann probes. As Bostrom gives examples of the possible developments he also lays it bare that motivations of a super intelligence might not align with humans, and furthermore if it could be aligned - with who. As a reader, my immediate thought is that most organizations that have the capital to develop such a technology might not be very willing to compromise with their vision, whatever it may be. Another point Bostrom makes is that software might be very labour and cost intensive, but replication of software is not. Therefore a general access to super intelligence might be of greater danger than a closed off one. Also returning to the core of defining what a super intelligence is, can it be non-autonomous and fully 'intelligent'?

The ethical dimensions surrounding the possible developments are alarming as well, for me this raises questions as to what we should and shouldn't do as a species. Is there a direct need for this, who does it serve? Bostrom further raises the point that it is not a given that human life should be placed above any other, we might even be competing with the continuation of the planet earth itself.

"The implausibility of intelligence explosion" by Chollet (2018) was a great read following Bostrom's article. Chollet questions what intelligence truly is and encapsulates the thought by context shifting the human intelligence both in time and space. What could or would happen if Einstein's brain was indeed placed in a farmer's hand's cranium, could we expect the human he became to be known as? This argument is made in the context of a dissection of the implausibility of an intelligence explosion, what is intelligence to begin with, and where are bottlenecks that could hinder the exponential growth and instead turn it linear? Chollet argues that intelligence is made by interactions and context, an AI cannot simply recursively progress through only the rewriting of code. While reading his arguments for what intelligence is and what it could be in the future, I find myself not completely convinced, the implication is that a super intelligence should be sentient, be able to experience in the physical world to be intelligent. And just as this might be true overall, in that a program can only know the color 'red' by information and not as a knowledge, this color is not part of some being's seeing spectrum, so on and so forth with many of our senses. What does intelligence actually boil down to?

Chollet's article was a very interesting read in that we come to question if super intelligence could come to exist, whilst Bostrom fills us with both trepidation and excitement for such an occurrence. These text have given me further reason to be wary of an unregulated rise of super intelligence, but at the same time understand that it were to become a reality, no amount of regulation can control a true super intelligence - if it truly is intelligent.

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Bahar Kimanos, bkimanos@kth.se

Texts: 5 and 6

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In *What Worries me about AI*, François Chollet starts writing about how people are always worried about new technological inventions, one of his examples being the fear of potential technological unemployment that people thought that the introduction of the internet would bring (Chollet, 2018). He states that even if people feared being replaced by computers, most of us have now learned to work with it, even the people who once hated it, and the fears we had are mostly long gone (Chollet, 2018). Instead there are many problems with computers and the internet today that weren't discussed back in the 80s and 90s, such as hackers, alienation and mental health issues due to social media, loss of focus and patience, mass surveillance and the fast spreading of political and religious propaganda (Chollet, 2018). We weren't knowledgeable about what new issues the internet would bring, and Chollet explains how our initial fears of new technology might later on turn out to be irrational, and then the real worrying developments are revealed after the technological change has already rooted itself into our lives (Chollet, 2018).

Based on these observations, Chollet states that what really worries him is the effective and scalable manipulation of human behaviour that AI enables, and how it can be used maliciously by corporations, governments and other bad actors (Chollet, 2018). This paper was published in 2018 and we can already see that his fears have turned out to be true. Multiple malicious actors have generated false imagery of big events worldwide to spread misinformation (Nilsson & Sköld, 2026) and corporations have trained their AI systems to perform tasks without proper safeguards that have been abused to create deepfakes (Andersson, 2026) and child sexual assault material (Vallance, 2026).

In *The Real Risks of Artificial Intelligence*, David Lorge Parnas mentions how "AI" is a very broad label with no generally accepted definition (Parnas, 2017). This term can cover anything from a system that is using very sound logic or a very sloppily built system with a lot of flaws (Parnas, 2017). He states that if we learn to depend on these programs labelled "AI", we might not have the capability to proceed without the programs input, even if it might be incorrect (Parnas, 2017).

Based on these two papers and news covering the recent developments of AI, François Chollet and David Lorge Parnas's fears might've come true already. For a few years, everything was labelled as "AI" without there being a proper definition of what that meant, and there still isn't. People blindly trust services because they're labelled as "intelligent", meanwhile they spread misinformation and are used by malicious users to cause harm to others. Capitalism often encourages companies to develop addictive algorithms and tools that can easily be abused as long as there aren't any legal regulations around them since this

attracts and keeps people using their services and companies don't experience legal ramifications. There have been recent discussions and developments regarding AI regulation and legislation (*AI Act*, 2026), but a lot of damage has been done and there's still a lot that needs to be done. Without proper regulations, AI will be abused.

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Mo Kong, mkong@kth.se

Texts: 10 and 14

Word Count: 549

Artificial intelligence can decode genomes, predict diseases, and personalise treatments, but doing so requires massive amounts of our most intimate data. The core question is not whether to use AI, but how to protect privacy and data security. Reading Pauwels and Vidyarthi (2017) alongside Müller (2017), I argue that legal compliance is merely a baseline. Genuine protection demands explicit ethical principles, institutional reform, and technical safeguards that go beyond what the law currently requires.

Pauwels and Vidyarthi vividly show why privacy and security cannot be an afterthought in the AI-genomics race. They warn that biological data are treated as “the new coal, oil or shale gas” (2017, p. 5), fuelling a billion-dollar industry where firms like iCarbonX aim to sequence 100 million genomes. However, this gold rush occurs on fragile cyberinfrastructure. Major breaches—at Anthem, the Office of Personnel Management, and Quest Diagnostics’ genomic platform—exposed tens of thousands of patients’ records (Pauwels & Vidyarthi, 2017, p. 5). Moreover, existing laws such as HIPAA and GINA were never designed for an AI-powered health economy, where algorithms can infer far more than legislators anticipated (p. 6). I agree completely: we are building a data-intensive future on legal foundations suited for a less connected era. Security flaws are not minor technical issues; they threaten public trust itself.

Müller’s conceptual distinction between legal obligations and ethical demands explains why legal compliance is insufficient. He argues that the EPSRC robotics principles conflate the two, treating what is already mandatory as optional and failing to articulate additional ethical standards (2017, p. 2). Applied to data protection, a company may comply with HIPAA’s consent requirements yet still use patient data in ways patients would find unethical—for example, selling predictive health insights to insurers without transparency or benefit-sharing. As Müller notes, “if we accept an ethical rule, this does not imply that there is or should be a legal rule” (p. 3), but conversely, legality does not imply ethical adequacy. I fully endorse this view. In AI and genomics, we cannot rely on regulatory checklists alone. We need proactive ethical principles: algorithmic transparency, meaningful user control over secondary data uses, and independent audits of how AI models infer sensitive attributes.

Combining both perspectives points toward actionable solutions. Pauwels and Vidyarthi recommend interdisciplinary “brain trusts” to anticipate governance challenges (p. 10) and new models of personal data ownership (p. 6). Müller insists that ethical demands must be grounded in theory and stated clearly as exceeding legal minima (p. 5). I find these proposals persuasive, but I would add that we must also invest in privacy-preserving technologies—federated learning and differential privacy, for instance—which

allow AI to learn without directly accessing raw genomic sequences. These tools should be treated as ethical imperatives, not optional features.

In conclusion, protecting privacy in the age of AI is not primarily a legal drafting exercise; it is a governance and design challenge. The U.S.-China competition described by Pauwels and Vidyarthi creates perverse incentives to cut corners on security and consent, while Müller shows that ethical guidelines must reject the logic of “just enough compliance.” The most important takeaway is that we must embed privacy and security into the very architecture of AI systems, foster international dialogue on shared norms, and always remember that our sensitive data should serve human flourishing—not corporate profits or geopolitical advantage.

References

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Judy Florence, judyf@kth.se

Texts: 8 and 13

Wordcount: 570 words

A big part of both papers that I read was ethics. When an AI causes harm, who should be held responsible? Even though both papers were written before today's generative AI became widespread, the philosophical questions they raise are still very relevant. The main takeaway from both papers, in my opinion, is that ethical decisions cannot simply be understood by looking at the outcome. There are a lot of things that need to be taken into consideration, such as intention, responsibility, agency, and the way

the harm is caused.

In “When HAL Kills, Who’s to Blame?”, Dennett uses the fictional computer HAL to question whether an intelligent computer could itself be morally responsible for its actions. HAL appears to have beliefs, goals, and the ability to make decisions, which would lead many to treat him as a sentient being. However, this raises an important question: if an AI makes a harmful decision, should we blame the AI, its programmers, the company that developed it, or the people who deployed it?

This becomes even more interesting when looking at some of the newer cases involving LLMs over the past few years. One case that comes to mind is a story where an AI chatbot reportedly reinforced a man’s belief that his mother was a spy, which eventually contributed to him killing her and then himself (Solis, 2025). It is interesting to see how the topics and questions raised by Dennett have only become more relevant as technology and AI have developed.

I think this is especially true when it comes to the question of who is to blame. Back when computers were mostly based on simpler algorithms, it was easier to assign responsibility to the people who created them. Nowadays, with the rise of large language models, things are much more complicated. I would still place the responsibility mainly on the creators and companies behind these systems, but I think it is harder to directly blame them for every single outcome. LLMs are extremely complicated systems, and even their creators do not fully understand why they produce every individual response. This makes the question of responsibility much harder to answer.

Thomson’s “The Trolley Problem” gives another interesting perspective. Her famous examples show that two actions can have similar outcomes but still feel morally different. Pulling a lever to redirect a trolley, killing one person instead of five, seems more acceptable to many people than pushing a person off a bridge to stop the trolley. The difference suggests that ethical decisions are not only about calculating consequences.

A system that only focuses on the outcome might simply choose the option that saves the most people. But Thomson’s argument shows that this might not always be enough. Things like rights, dignity, intention, and how the decision is made also need to be considered.

My main takeaway from these readings is that AI ethics should not only focus on making machines more intelligent. We also need to think about how we can make them act in ways that fit our ethical values. I do think that would be extremely difficult, though, since even humans disagree about what is right and wrong and what the best solution is. Because of this, I cannot imagine a near future where we would completely put decisions involving human lives in the hands of machines. At least for now, I think humans still need to have the final say.

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Summary text

Bahar's texts were about the risks of AI and what the writers feared could be problems in the future. The group discussed how interesting it was seeing that the papers from 2017 and 2018 so accurately predicted problems that would arise with the publicly accessible AI chatbots. We all brought up different situations that we had read about, mainly regarding ChatGPT and Grok, and how they had played a role in people's worsened mental health issues and deaths. The papers mentioned how new technology often brings along new terms and problems that were hard to predict, similarly how these AI chatbots have brought along new problems. Since everyone in the group had already read about ways AI had harmed people, nobody was really surprised and the conclusion before and after the discussion was that AI being so available to the general public is problematic. Even if they put up safeguards, new issues arise and it's hard to predict what the next problem is going to be.

Mo's first article examines the U.S. China competition in AI-driven genomics, highlighting how data breaches could expose individuals' genetic information and undermine privacy. The second article critiques the EPSRC robotics principles, arguing that they conflate legal obligations with ethical demands and merely restate existing legal requirements without articulating additional ethical standards. The author recommends revising the principles to clearly distinguish legal duties across different actors, while emphasizing that human responsibility persists regardless of robot autonomy. Many of these recommendations, such as clearer legal-ethical distinctions, remain relevant today, though their full implementation is still ongoing.

Nor's first essay is arguing that intelligence isn't something you can grow by incursion. You can't take Einstein's brain and put it in the body of a farmer from 2000 years ago and they won't become super intelligent because intelligence is context and time based. But is this really true, does intelligence include the experience of colors, or is the knowledge of it conclusive? We humans aren't able to see the full color spectrum, are we somewhat lacking in intelligence because we lack the experience? Or maybe that's a weak conclusion, instead we can say that not experiencing those colors has not led to 'development' in regard to that.

The second paper was about how we as engineers can program AI to be human friendly, but if AI is truly going to be intelligent, humans wouldn't be able to control them, and why would they care about humans and their safety? This sparked a discussion on the sentience of AI, what is the difference between being sentient and not - is it relevant to what choices AIs make? What is it that makes humans sentient? Doesn't superintelligence equate to sentience? And would that sentience be the end of human supremacy? Judy thinks that if something acts sentient, then it should be considered sentient. The discussion turned to how 'sentience' is an elusive concept, many believe sentience might be of the metaphysical, which would exclude AI in most people's understanding.

Judy's text was similar to Nor's, and we continued discussing the question of AI sentience and whether AI could ever be considered similar to humans in this way. Judy points out that, as humans, we are heavily influenced by our surroundings and by the information we are exposed to. This led us to wonder whether this is actually that different from how AI works. AI models are also influenced by the data they are trained on and by the information they receive from users.

Bahar thought there was still an important difference between humans and AI, but she found it difficult to explain exactly what that difference is. We discussed whether the difference could be related to consciousness or the fact that humans actually experience the world around them, while AI only processes information about the world. This brought us back to the question of whether being influenced by our surroundings is enough to make something sentient, or if there needs to be some kind of personal experience or awareness behind it.